

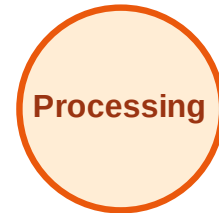
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing

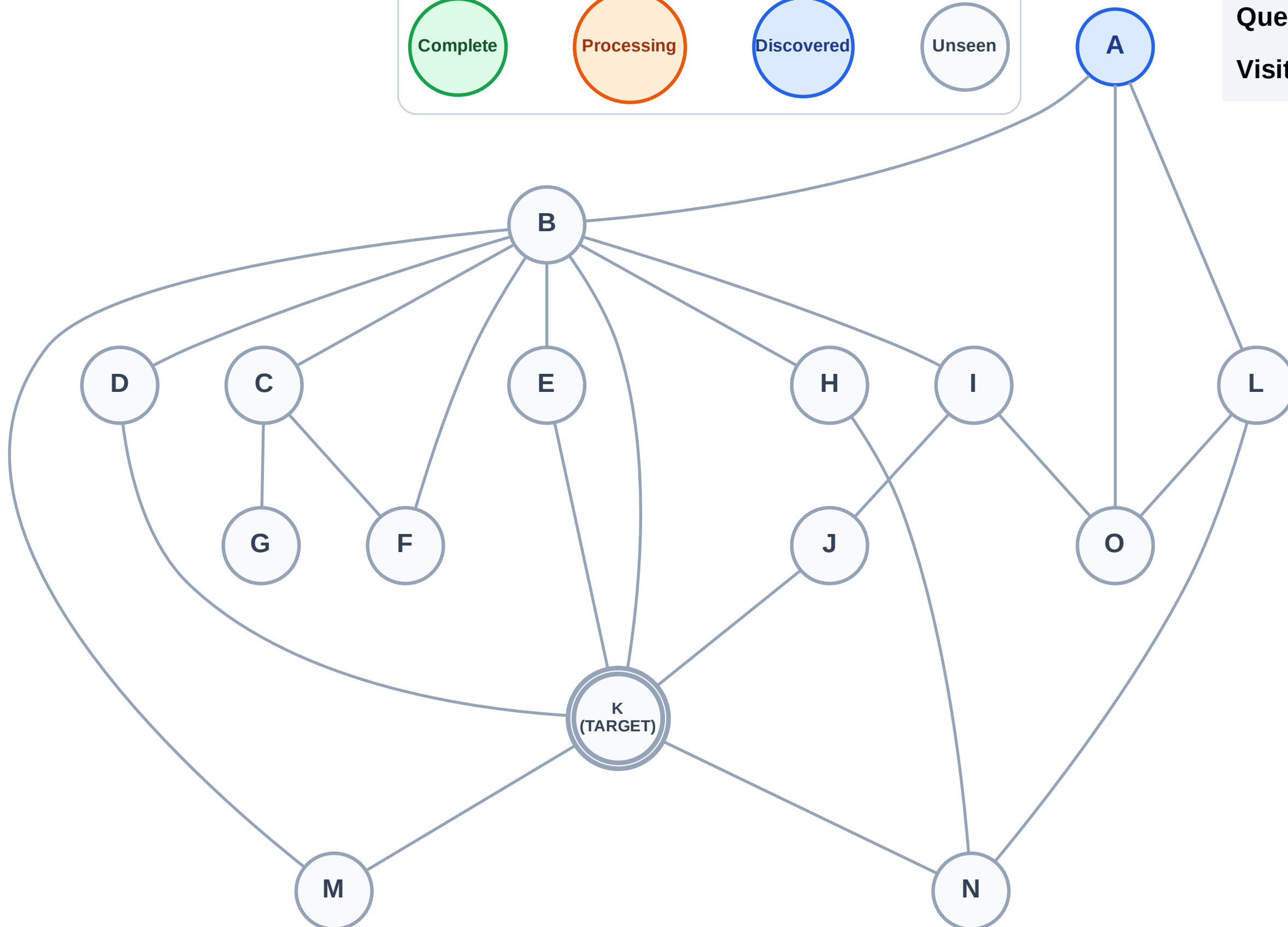


Discovered



Unseen

Queue: A
Visited: (none)



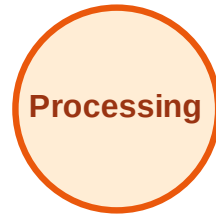
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



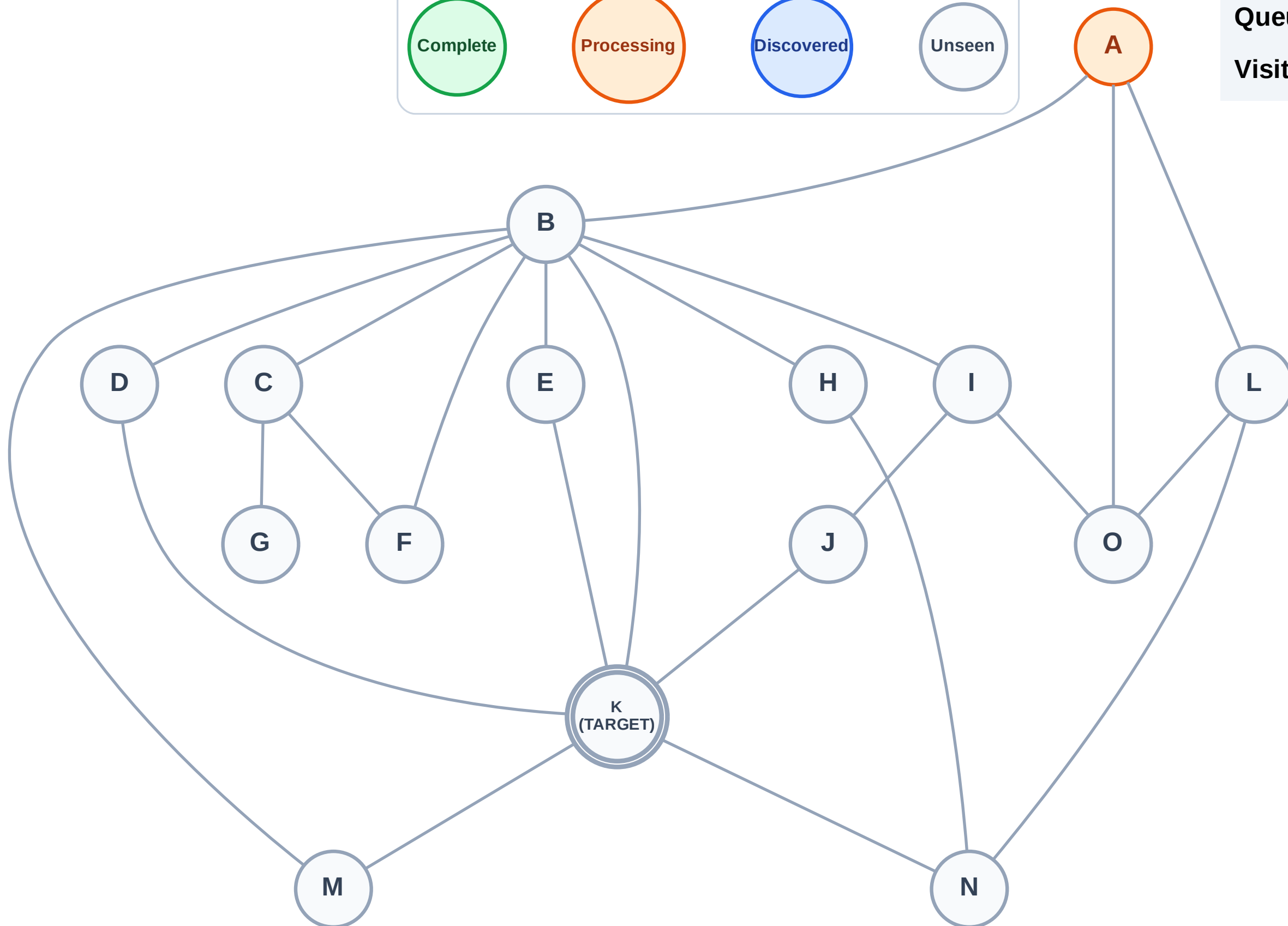
Discovered



Unseen

Queue: (empty)

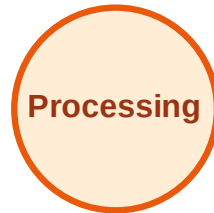
Visited: (none)



BFS Traversal

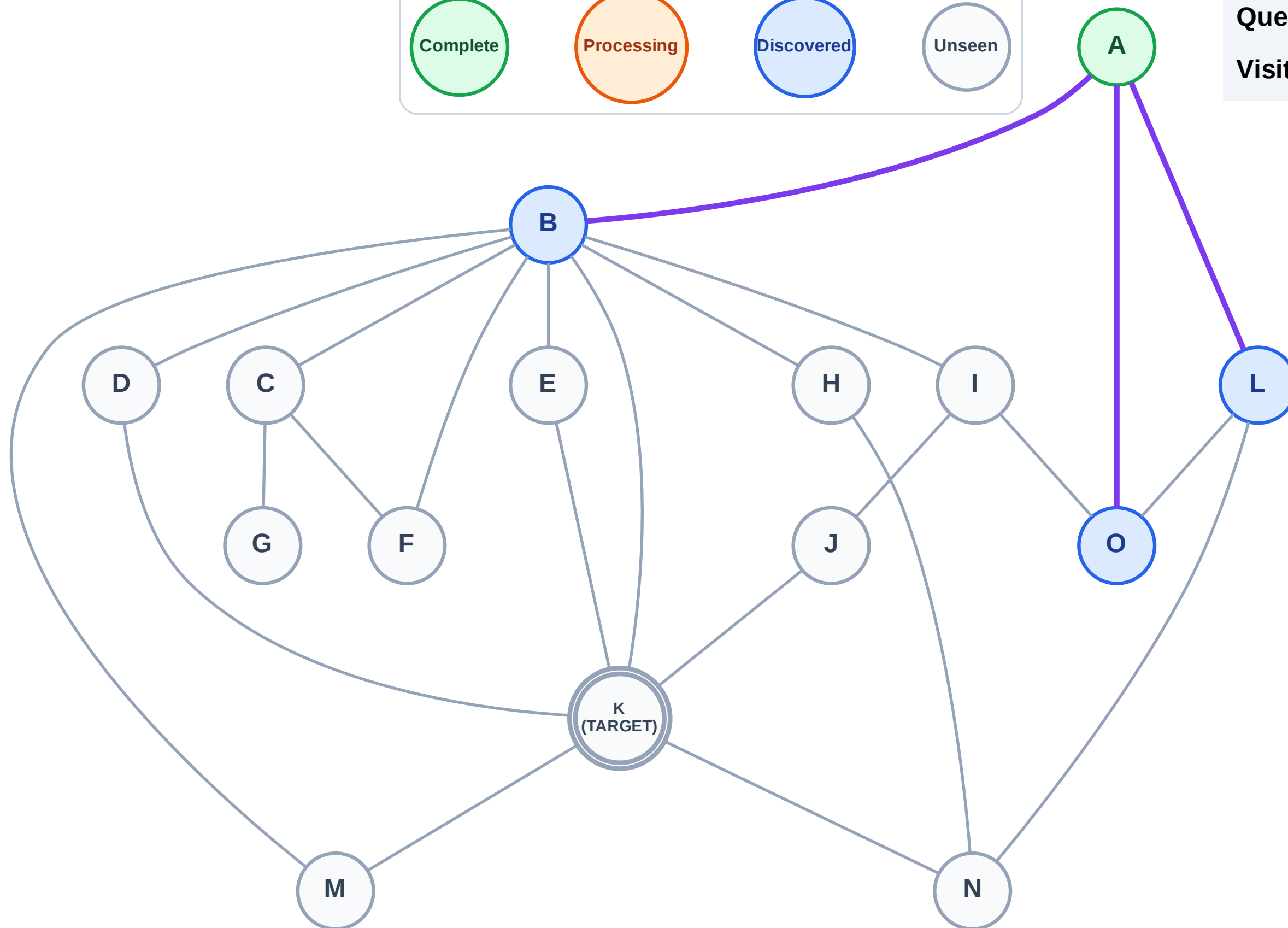
Step 3: Discover B, L, O from A

Legend



Queue: B → L → O

Visited: A



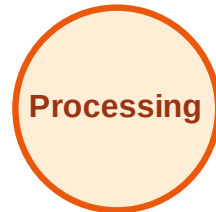
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



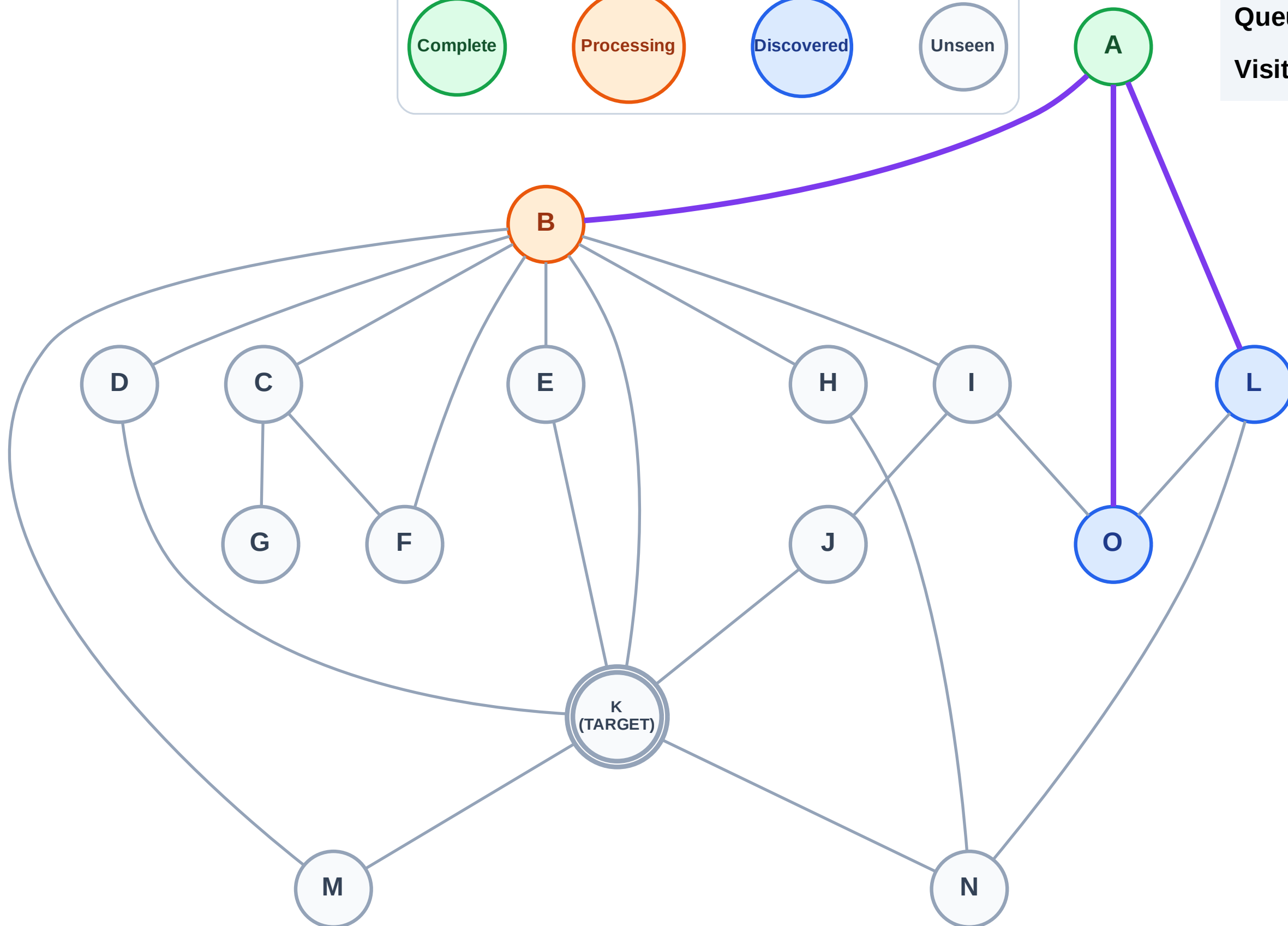
Discovered



Unseen

Queue: L → O

Visited: A



BFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend

Complete

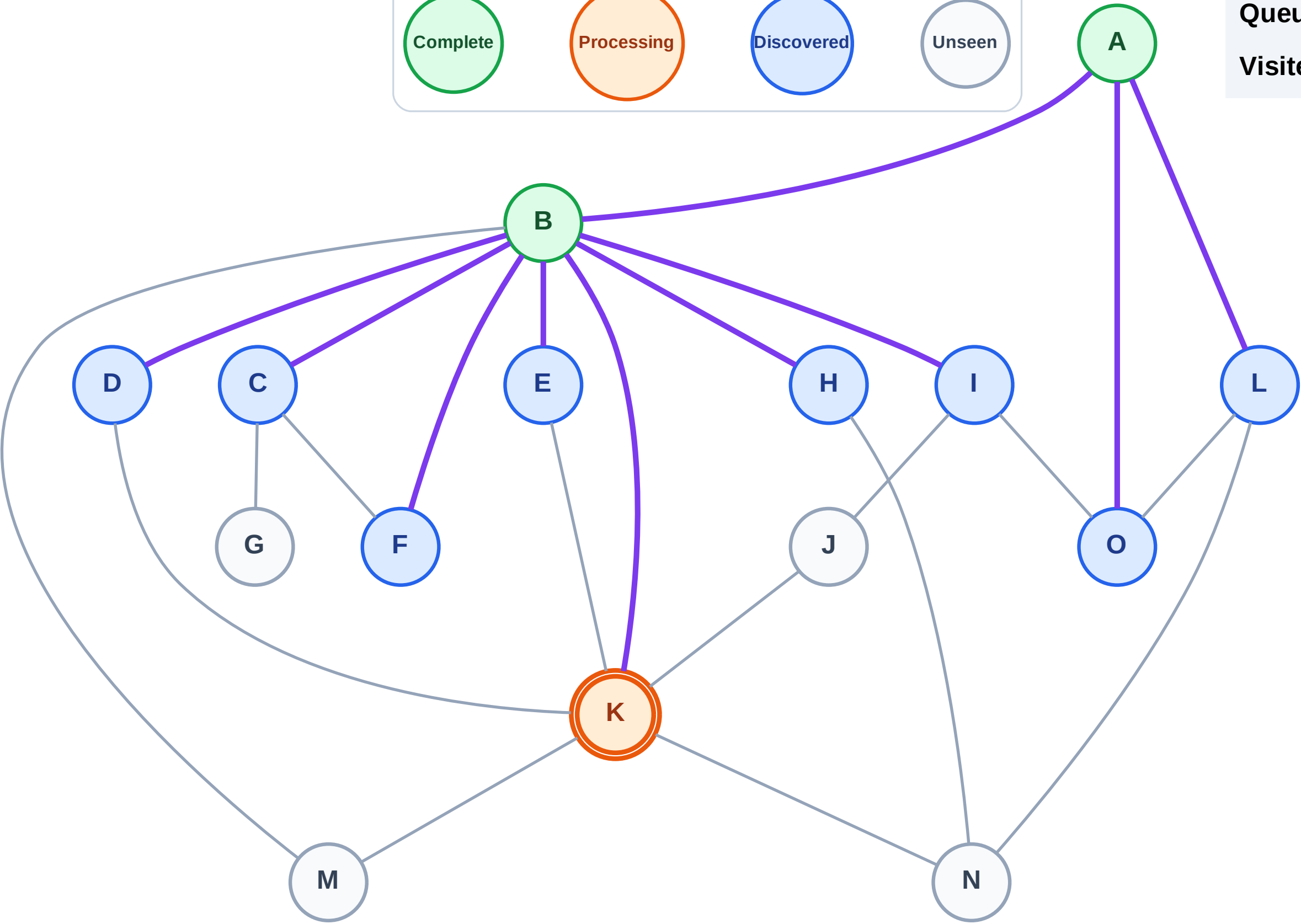
Processing

Discovered

Unseen

Queue: L → O → C → D → E → F → H → I

Visited: A, B, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

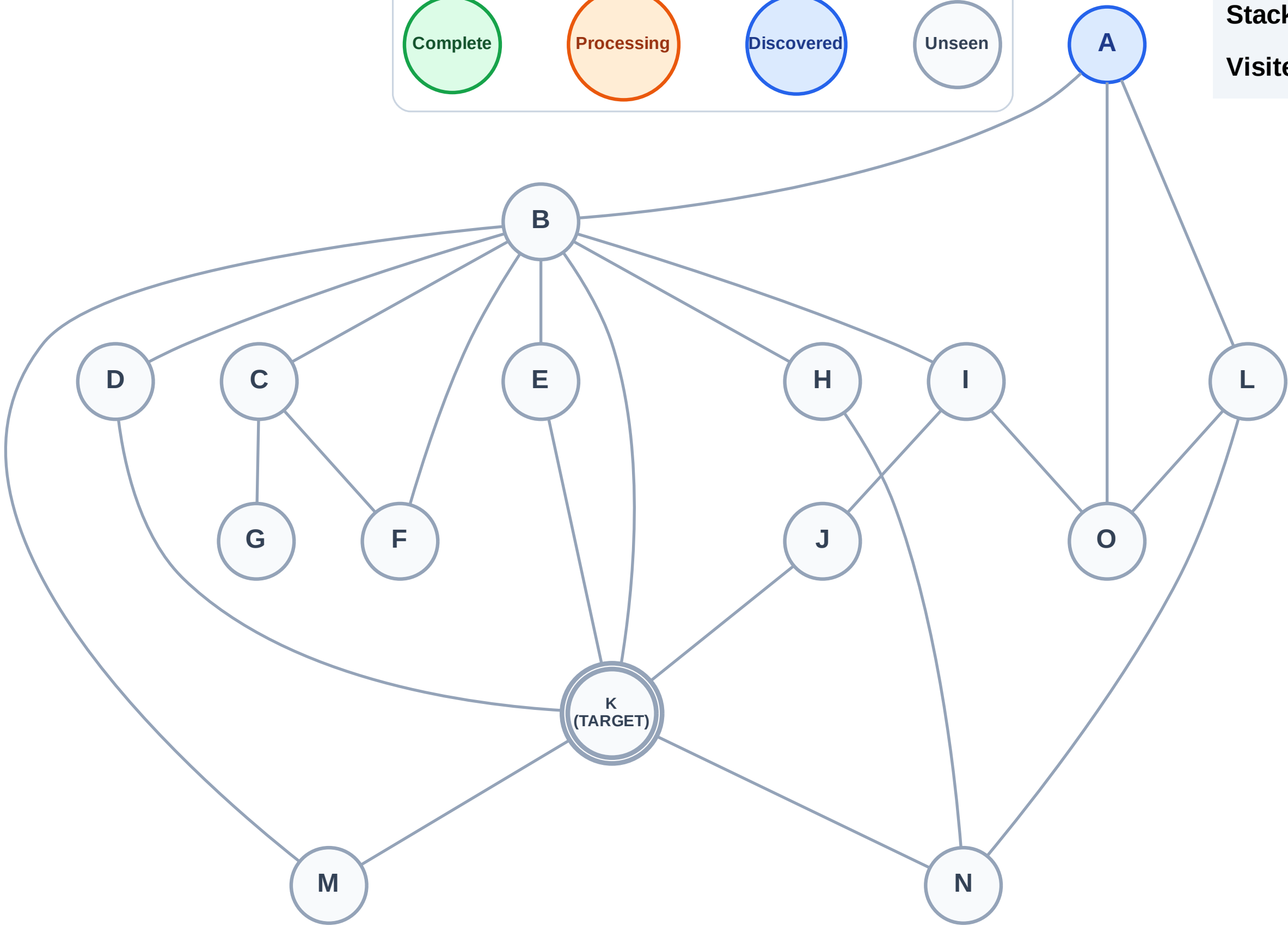
Complete

Processing

Discovered

Unseen

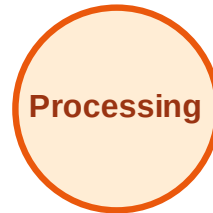
Stack (top at right): A
Visited: (none)



DFS Traversal

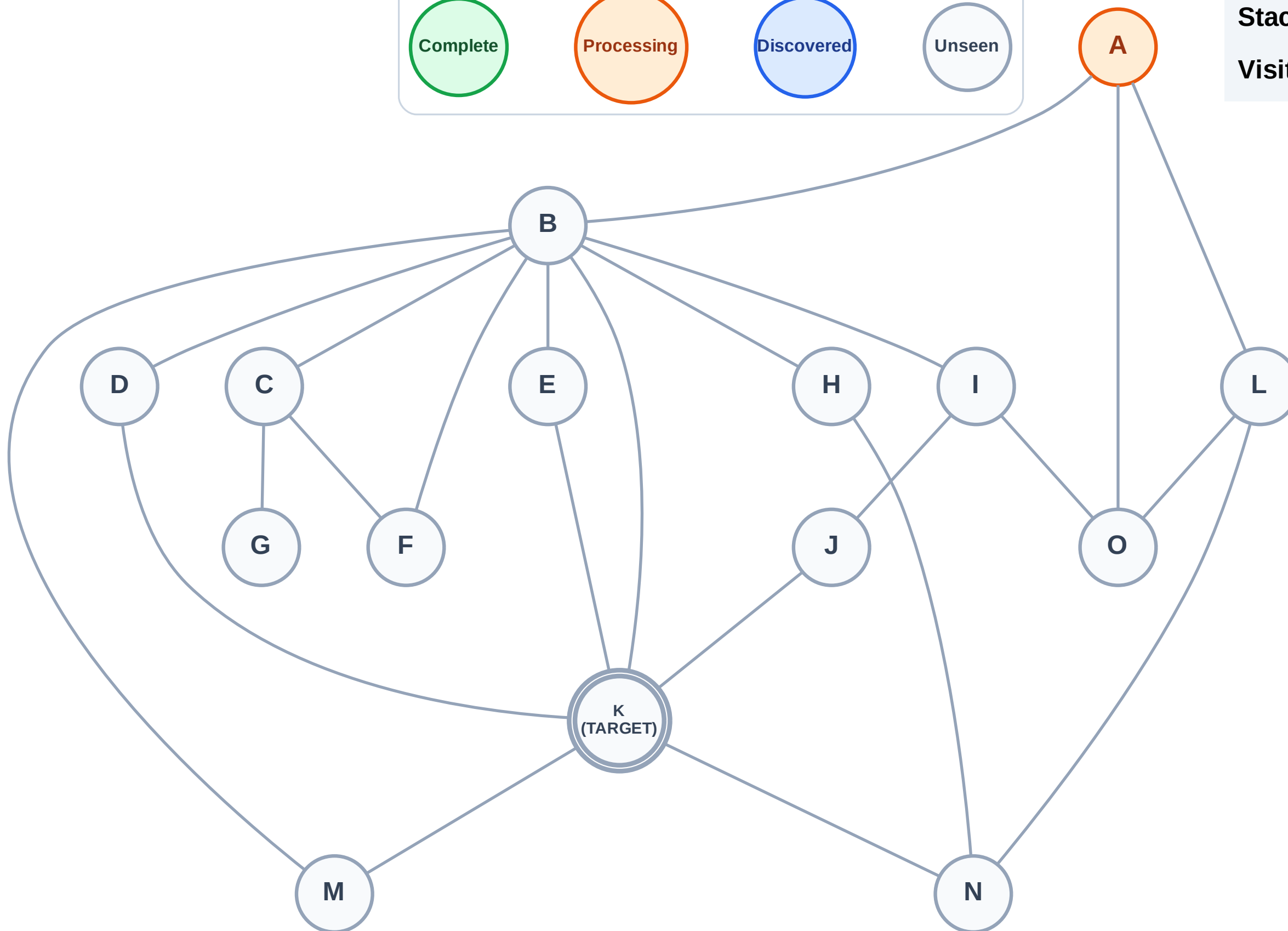
Step 2: Process node A

Legend



Stack (top at right): (empty)

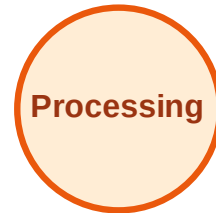
Visited: (none)



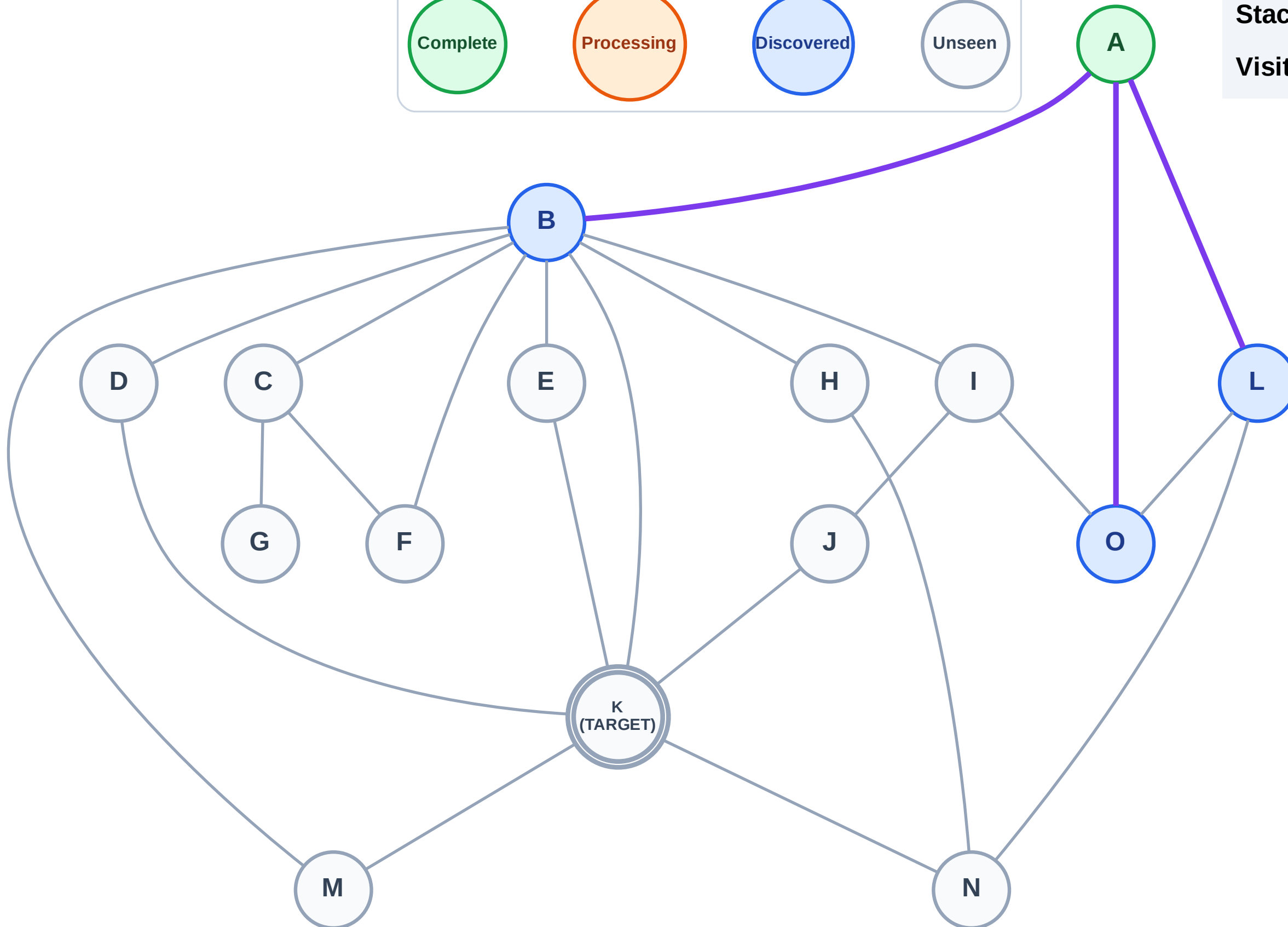
DFS Traversal

Step 3: Discover O, L, B from A

Legend



Stack (top at right): O | L | B
Visited: A



DFS Traversal

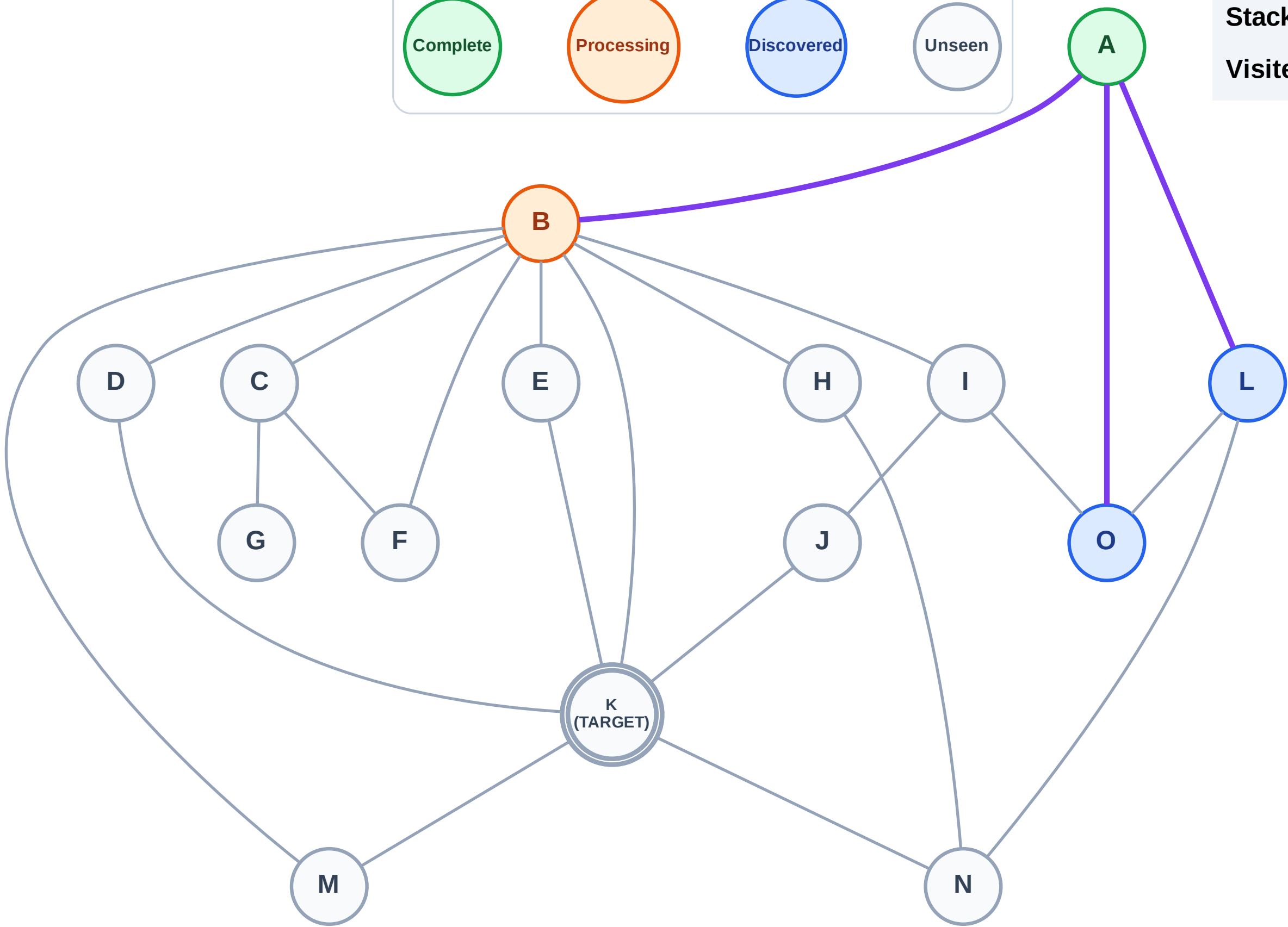
Step 4: Process node B

Legend



Stack (top at right): O | L

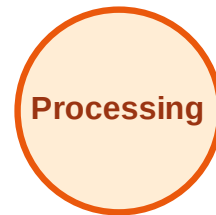
Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Stack (top at right): O | L | M

Visited: A, B, K

